LIVE CLOUD PROJECT

Enterprise Web Portal

Micro Group Institute

Smart Learning, CBT Exam & Automated Certification Portal

An end-to-end cloud platform for computer educational institutes featuring strict anti-cheat online exams, automated grade evaluations, typing speed analytics, and verified digital certificates.

Anti-Cheat Exam
Auto-submit & tab control

Instant Certificate
Unique online verification ID

Typing Test Center
WPM & accuracy tracking

Cloud PostgreSQL
Dockerized & persistent

Speaker Note: "Hello everyone. Today I am presenting our live cloud platform for Micro Group Institute. Unlike standard static websites, our system completely automates the institute's examination, typing tests, and certification process."

THE PROBLEM

Challenges in Traditional Institutes

Why traditional paper exams and basic websites fail modern computer education.

Paper Waste & Slow Results

Printing question papers is expensive. Teachers spend days manually grading answers, causing delayed results.

Rampant Exam Cheating

Online quizzes like Google Forms allow students to open new tabs, Google answers, and copy-paste easily.

Fake & Unverifiable Certificates

Paper certificates are easily forged. Employers have no fast online way to verify if a student's diploma is authentic.

Speaker Note: "Most computer institutes still print paper exams or use Google Forms where students cheat by opening Google. Furthermore, paper certificates can easily be duplicated without any verification."

THE SOLUTION

How Our Portal Solves This

A direct comparison between standard websites and our automated portal.

Feature	Typical Institute Website	Our System (Micro Group Portal)
Website Nature	⊗ Static Pages (Visiting Card only)	✓ Complete Web App & Examination ERP
Online Exam Integrity	⊗ Easy to cheat via Google / Tabs	✓ Anti-Cheat tab detection & auto-submit
Result Generation	⊗ Manual checking (Takes 3-5 days)	✓ Instant score, grade & pass/fail in 1 sec
Certificate Authenticity	⊗ Offline paper (Unverifiable)	✓ Unique Certificate ID with public portal verification
Typing Test System	⊗ Requires desktop software installation	✓ 100% in-browser WPM & accuracy engine

Speaker Note: "As shown on screen, while other websites act only as static digital brochures, our platform completely handles the student's lifecycle: from study material to cheat-free testing and verified certification."

CORE INNOVATION

Intelligent Anti-Cheat Exam Engine

Ensuring 100% fair and honest testing on mobile phones and computers.

Tab-Switch Detection

Monitors screen visibility using the Page Visibility API. If a student switches to Google or WhatsApp, an immediate alert triggers.

3-Strike Auto Submit

Students receive warnings on suspicious activity. On the 3rd attempt, the exam auto-submits and terminates immediately.

Copy / Paste Lock

Right-click, text selection, inspect element (F12), and keyboard shortcuts (Ctrl+C, Ctrl+V) are strictly blocked.

Speaker Note: "Our examination engine includes a proprietary client-side security layer. Even if students test on mobile phones, switching apps or copying questions is prevented, making the examination 100% credible."

INSTANT VALIDATION

Instant Results & Certificate Verification

Empowering students with instant feedback and verifiable credentials.

Real-Time Automated Grading

The moment a student submits their exam:

- Calculates correct answers, wrong answers, and marks.
- Computes exact percentage and automated grade (A+, A, B, etc.).
- Stores the complete historical attempt safely in PostgreSQL.

Public Verification Engine

Institutes issue authentic credentials:

- Auto-generates standardized IDs (e.g., MGI-2026-00001).
- Employers or schools enter the Roll No or Certificate No.
- Instantly renders student name, course, score, and issue date.

Speaker Note: "Once the exam concludes, the student receives their grade instantly. Furthermore, any employer can visit our website's 'Verify Certificate' portal to authenticate student diplomas in seconds."

SKILL ASSESSMENT

Interactive Typing Test Module

A native browser typing tutor and assessment engine without third-party software.

Net WPM

Measures pure speed after deducting penalties for typing mistakes.

Accuracy %

Precise percentage of correct keystrokes vs total keystrokes.

Mistake Count

Tracks skipped words and typographical errors in real-time.

Typing Certificate

Issues instant typing assessment certificates with verified scores.

Speaker Note: "In computer education, typing is a vital skill. We built a native typing engine directly inside the browser. It tracks words-per-minute, errors, and accuracy in real-time."

SYSTEM ARCHITECTURE

Enterprise Tech Stack & Cloud Infrastructure

Built with industry-standard, lightweight, and scalable technologies.

Backend: PHP 8.2 & PDO

Object-oriented database queries using PDO Prepared Statements. 100% protected against SQL Injection and XSS attacks.

Database: Cloud PostgreSQL

Hosted on high-speed managed cloud databases. Fully persistent storage ensuring student records are never lost.

DevOps: Docker & Render

Containerized using custom Dockerfile with Apache & libpq drivers. Live deployed with automatic SSL/HTTPS encryption.

Speaker Note: "On the technical side, the application is built with modern PHP 8.2, containerized with Docker, and connected to an enterprise PostgreSQL database on the cloud."

LIVE SYSTEM READY

Project Summary & Demonstration

Experience the portal live in production.

🌐 Live Portal URL

`https://micro-group-institute.onrender.com`

- Mobile & Desktop Optimized
- 25-Question ADCA Challenge Live
- Full Admin Panel with RBAC

Thank You!

Ready for Questions & Answers (Q&A).

> **Speaker Note:** "Thank you so much. I will now open our live website to demonstrate the student exam process, anti-cheat protection, and instant certificate generation. I welcome any questions!"